

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
END SEMESTER THEORY EXAMINATION

Question Paper

Month and year: **July/August 2026**

Programme: **B.Tech**

Subject: **Computer Networks**

Maximum Marks: **50**

No. of pages: **1**

Semester: **4th**

Course code: **CSPC- 202&ITPC- 202**

Time allowed: **03 Hours**

Note1: Attempt **five** questions. Question **no-1** is compulsory.

Note2: Unless stated otherwise, the symbols have their usual meanings in context with subject. Assume suitably and state, additional data required, if any. The Candidates, before starting to write the solutions, should please check the question Paper for any discrepancy, and also ensure that have been delivered the question paper of right **course no.** and right **subject title**.

Q-1.	a) What are the propagation time and the transmission time for a 2.5 KB (kilobyte) message (an email) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s. b) A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 μs, what is the minimum size of the frame? c) A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 250 frames per second? d) What is the Hamming distance for the codewords, d (10101, 10000) and compute? e) We need to send data at a 1-Mbps rate. What is the minimum required bandwidth, using a combination of 4B/5B and NRZ-I or Manchester coding?	2*5
Q-2.	With a neat block diagram, explain in detail about TCP/IP layered architecture model and its layer operations.	10
Q-3.	a) Draw a neat block diagram of UDP protocol and explain their components. Why does the TCP header have a header length field while the UDP header does not? b) Why is TCP called connection-oriented? Can you explain why we need four (or sometimes three) segments for connection termination in TCP?	5*2
Q-4.	a) Draw a neat block diagram of IPv4 datagram format and describe. Also, compare the fields in the main headers of IPv4 and IPv6. b) Discuss how subnetting works in IPv6? Suppose an ISP has given you the IPv6 address 2001:3200:1000::/48. If your requirement is for 16 subnets then find out and list all full length IP addresses.	5*2
Q-5.	Explain access control protocols in details. How does this impact the efficiency of the transmission medium?	10
Q-6.	a) Explain the working of ARP protocol. Describe how the ARP is useful in IP addressing. b) Discuss the file transfer protocol (FTP) using suitable diagrams.	5*2